

Software System Development - Monsoon 2026
Quiz 1 Key

Q1: Consider a table of Employees with a column bonus containing values: [100, 200, NULL]
What is the result of the following query? _____

1. **SELECT**
2. **COUNT(*),**
3. **COUNT(bonus),**
4. **SUM(CASE WHEN bonus <> 100 THEN 1 ELSE 0 END)**
5. **FROM Employees;**

Answer: 3, 2, 1

- COUNT(*) counts all rows (including the one with NULL), resulting in **3**.
- COUNT(bonus) counts only non-NULL values (100 and 200), resulting in **2**.
- SUM(CASE WHEN bonus <> 100 THEN 1 ELSE 0 END) evaluates to 1 only for the value 200. For 100, it evaluates to 0. For NULL, the comparison results in UNKNOWN, triggering the ELSE 0. The sum is **1**.

Q2: Given two tables: Department with dept_id: [1, 2, 3] & Project with lead_dept_id: [1, NULL]. What does the following query return? _____

1. **SELECT dept_id**
2. **FROM Department**
3. **WHERE dept_id NOT IN (SELECT lead_dept_id FROM Project);**

Answer: An empty set (zero rows).

- The subquery returns (1, NULL). The NOT IN operator behaves like an AND condition (dept_id <> 1 AND dept_id <> NULL). Since any comparison with NULL yields UNKNOWN, the WHERE clause is never TRUE, so no rows are returned.

Q3: In data loading strategies, which approach involves loading raw incoming records into an intermediate table with minimal constraints or indexes before performing transformations, validations, and batched insertions into production?

Answer: Staging table (or Staging Area).

Q4: In relational databases, an Estimated Execution Plan is generated before query execution, whereas an Actual Execution Plan is a report generated after the query execution completes. **[True or False]**

Answer: True

Q5: A fact table stores descriptive attributes providing context, whereas a dimension table stores quantitative, numerical measurements.

Answer: False (The definitions are reversed; Fact tables store quantitative measurements, while dimension tables store descriptive attributes.)

Q6: A database designer encounters a table containing a 3-way relationship (Agent, Supplier, Product) that could be incorrectly reconstructed if split naively. To guarantee exact reconstruction, the designer decomposes it into precise binary tables. Which normal form is the designer strictly fulfilling?

Answer: Fifth Normal Form (5NF) / Project-Join Normal Form (PJNF).

Q7: In a relational database, an existing user application is designed to operate on only a specific subset of attributes within a relation. When the database administrator adds new, unrelated attributes to that same relation, the user application continues to function perfectly without requiring any updates. According to the text, what specific principle is protecting this application?

Answer: Logical Data Independence.

Q8: Fill in the blanks

a. **BASE:** Basically Available, Soft state, Eventual consistency

- b. Relational databases follow **ACID** properties, whereas Unstructured databases follow **BASE** properties.
- c. Relational databases have **schema-on-write** (or rigid/predefined) schemas and Unstructured databases have **schema-on-read** (or flexible/dynamic) schemas.
- d. A **Semantic Layer** (or Data Mart) is a business-facing abstraction layer...
- e. A **deletion** anomaly occurs when it is impossible to delete certain information without losing some other, unrelated information.
- f. Example type of Unstructured database: **Document database** (e.g., MongoDB), Key-Value store, or Graph database.
- g. **ACID**: Atomicity, Consistency, Isolation, Durability.

Q9: Identity 3 errors from the NoSQL Query below and mention the correction. Each error carries one mark.

```
db.user.updateMany(  
  { year: { $lt 2022 } }  
  { $get:: { city: Hyderabad, branch:'M.Tech EEE' } },);
```

1. **Error:** Updatemany
 - **Correction:** It should be camelCase: updateMany
2. **Error:** { \$lt 2022 }
 - **Correction:** Missing a colon after the operator: { \$lt: 2022 }
3. **Error:** { \$get:: ... }
 - **Correction:** \$get:: is not a valid update operator, and the syntax is wrong. It should be \$set:. Additionally, string values like Hyderabad need quotes (e.g., \$set: { city: 'Hyderabad', ... }).
4. **Error:** Missing comma between the filter document and the update document.
 - **Correction:** Add a comma after { year: { \$lt: 2022 } }

Q10: Write a NoSQL query to count number of documents in 'student' collection (A)

Answer: db.student.countDocuments() or db.student.count()

Q10: Write a NoSQL query to retrieve all documents from 'course' collection (B)

Answer: db.course.find() or db.course.find({})

Q10: Write a NoSQL query to count number of documents in collection (C)

Answer: db.<collection_name>.countDocuments() or db.collection.countDocuments()

Q10: Write a NoSQL query to permanently delete the entire 'games' collection along with all the documents stored inside it. (D)

Answer: db.games.drop()

Q10: Write a NoSQL query to count number of documents in 'college' collection (E)

Answer: db.college.countDocuments() or db.college.count()